

Elanchezhian Muthukumar

Software Developer

✉ elanchezhian2712.m@gmail.com | [in LinkedIn](#) | [📁 Portfolio](#) | [🐙 GitHub](#) | ☎ [+91-9789382712](tel:+91-9789382712)

SKILLS

- **Frontend:** React.js, Next.js, HTML5, CSS3, Bootstrap, Framer Motion
- **Backend & APIs:** FastAPI, Django, Flask, RESTful APIs, Google Drive API, Gemini AI API
- **Databases:** MySQL, PostgreSQL, MongoDB, Supabase
- **Languages:** Python, JavaScript (ES6+)
- **Cloud & DevOps:** AWS, Google Cloud, Docker (Basic)
- **Tools:** Git, GitHub, Postman, Figma, Visily AI

EDUCATION

- **PERIYAR UNIVERSITY**
MASTER OF COMPUTER APPLICATIONS — University Rank Holder
Aug '22 — Jun '24 | CGPA: 8.5
- **VYSYA COLLEGE**
BACHELOR OF SCIENCE IN COMPUTER SCIENCE
Jul '19 — Jun '22 | CGPA: 7.7

CERTIFICATIONS

- Full Stack Web Developer Certification – Udemy (Jun '24)
- Java and Python Programming Certification – NPTEL (Feb '23)
- AI for India 2.0 Certification – GUIV, Skill India (Aug '24)
- Innovation, Design, and Entrepreneurship Bootcamp – MIC AICTE (Jun '23)

AWARDS & HONORS

- **1st Place**, Project Expo, Emerge 2K23, SRM Valliammai Engineering College (Nov '23)
- **Consolation Prize**, In-House Project Expo-2K23, Periyar University (Sep '23)

PROFESSIONAL EXPERIENCE

SOFTWARE DEVELOPER

MAHIMA TECHNOLOGY PRIVATE LIMITED | SALEM, TAMIL NADU, INDIA

July '24 — Aug '25

- **Automated key business processes** by engineering a user time-tracking system and integrating a Gemini AI pipeline for invoice data extraction, **reducing manual reporting time by 50%**.
- **Designed and deployed a dynamic file automation platform** to consolidate datasets, enforce validation, and trigger automated exception alerts via email, decreasing reconciliation effort by 40%.
- **Constructed a dual QC/QA PDF annotation tool** using Python and Django, enabling side-by-side comparison against design specifications and automated discrepancy tracking, improving early defect detection by 15%.

WEB DEVELOPMENT INTERN

RAKUMURA IT SOLUTIONS | CHENNAI, TAMIL NADU, INDIA

Jan '24 — Apr '24

- **Led the end-to-end development of responsive websites** for multiple clients, resulting in a **40% improvement in user engagement**.
- Collaborated with cross-functional teams to **launch 10+ new feature initiatives**, directly contributing to a **25% increase in quarterly sales**.

PROJECTS

SECURE AI DOCUMENT & BOOKMARK VAULT

TECHNOLOGIES: NEXT.JS, GEMINI AI, TESSERACT.JS OCR, CRYPTOJS, AES ENCRYPTION, GOOGLE DRIVE API | [\[LINK\]](#)

- **Implemented a zero-knowledge security model** using client-side AES encryption, Google OAuth 2.0, and two-factor authentication (2FA) to ensure complete data privacy and security.
- **Developed a client-side Optical Character Recognition (OCR) pipeline** with Tesseract.js, automating data extraction from over 800 documents daily with 96% accuracy.
- **Integrated Google Gemini AI for Natural Language Processing (NLP)**, enabling voice and text commands for vault management and reducing user task time by 2.5 hours per week.

VIRTUAL ASSISTANT FOR INDIVIDUALS WITH DISABILITIES

TECHNOLOGIES: PYTHON, FLASK, REST APIs, JAVASCRIPT, HTML5, CSS3 | [\[LINK\]](#)

- **Developed an AI-powered desktop assistant** with a speech-to-text engine to provide hands-free system control, improving computer interaction efficiency by 50% for users with disabilities.
- **Engineered a modular Flask backend and REST API** to support intelligent image and speech processing modules for rapid, reliable task automation.
- **Validated system effectiveness** through user testing with 10+ participants, confirming a 50% improvement in interaction efficiency and enabling independent computer use.

EMPLOYEE MANAGEMENT SYSTEM

TECHNOLOGIES: MONGODB, EXPRESS.JS, REACT.JS, NODE.JS, JWT | [\[LINK\]](#)

- **Secured the application with Role-Based Access Control (RBAC)** for 3 user tiers (Admin, Manager, Employee), protecting all sensitive HR data.
- **Built a high-performance REST API** using JWT for authentication, managing 20+ endpoints with an average response time of under 150ms.
- **Reduced administrative workload by 40%** by developing a self-service dashboard that automated leave and attendance requests.